

FEATURE AEROSPACE

How the Boeing 737 Max Disaster Looks to a Software Developer

Design shortcuts meant to make a new plane seem like an old, familiar one are to blame

BY GREGORY TRAVIS



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This is part of the wreckage of Ethiopian Airlines Flight ET302, a Boeing 737 Max airliner that crashed on 11 March in Bishoftu, Ethiopia, killing all 157 passengers and crew. PHOTO: JEMAL COUNTLESS/GETTY IMAGES

U **PD**ATE: 3 FEB. 2024: IN THE WAKE OF A NEAR-disastrous cabin blowout in an Alaska Airlines 737 Max 9 on 5 Jan.—which has propelled the embattled redesign of the once iconic 737 back into the headlines—Boeing, the plane’s designer, has been shaken anew. As the *Financial Times* reported on 31 Jan., the company’s chief executive Dave Calhoun wrote in a note to employees that

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accountable for what happened. ... we simply must be better. Which perhaps explains why the company even declined to report their financial outlook for the remainder of 2024.

In response to the company's near-crisis state, *The Los Angeles Times* reported on 30 Jan. that a former Boeing manager has publicly stated his aversion to ever flying on the redesigned 737. "I would absolutely not fly a Max airplane," former Boeing senior manager Ed Pierson told the paper. "I've worked in the factory where they were built, and I saw the pressure employees were under to rush the planes out the door. I tried to get them to shut down before the first crash."

As Gregory Travis chronicles below from a 2019 perspective, a latter-day crisis of faith about the "Max" appears to be building even more in size and stature. On 1 Feb., Reuters reported that demand for older 737 jetliners has skyrocketed since 2024's near-disaster. As Travis describes it, the original, classic craft with "smallish engines and relatively simple systems" is, even in its advanced age, proving very tough to

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I have been a pilot for 30 years, a software developer for more than 40. I have written extensively about both aviation and software engineering. Now it's time for me to write about both together.

The Boeing 737 Max has been in the news because of two crashes, practically back to back and involving brand new airplanes. In an industry that relies more than anything on the appearance of total control, total safety, these two crashes pose as close to an existential risk as you can get. Though airliner passenger death rates have fallen over the decades, that achievement is no reason for complacency.

The 737 first appeared in 1967, when I was 3 years old. Back then it was a smallish aircraft with smallish engines and relatively simple systems. Airlines (especially Southwest) loved it because of its simplicity, reliability, and flexibility. Not to mention the fact that it could be flown by a two-person cockpit crew—as opposed to the three or four of previous

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both for the industries that make them and the customers who buy them, and they all go through a similar growth process.

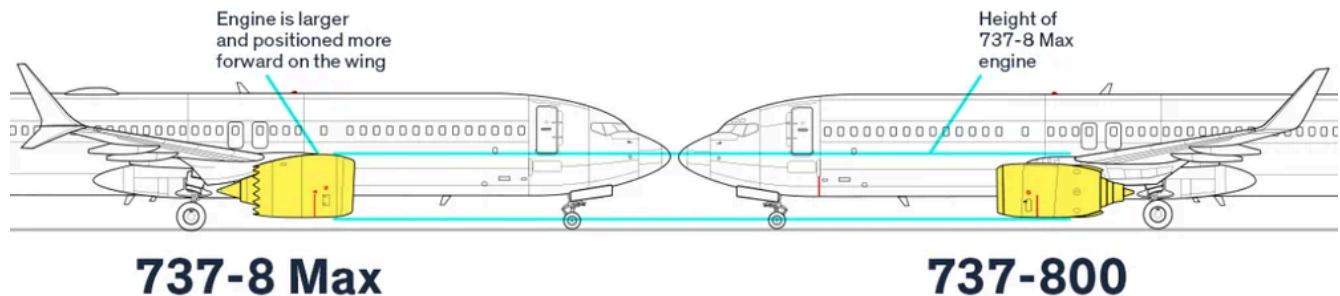
Most of those market and technical forces are on the side of economics, not safety. They work as allies to relentlessly drive down what the industry calls “ seat-mile costs”—the cost of flying a seat from one point to another.

Much had to do with the engines themselves. The principle of Carnot efficiency dictates that the larger and hotter you can make any heat engine, the more efficient it becomes. That’s as true for jet engines as it is for chainsaw engines.

It’s as simple as that. The most effective way to make an engine use less fuel per unit of power produced is to make it larger. That’s why the Lycoming O-360 engine in my Cessna has pistons the size of dinner plates. That’s why marine diesel engines stand three stories tall. And that’s why Boeing wanted to put the huge CFM International LEAP engine in its latest

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with bigger engines, the clearance between the engines and the ground started to get a little...um, tight.



By substituting a larger engine, Boeing changed the intrinsic aerodynamic nature of the 737 airliner. NOREBB0.COM

Various hacks (as we would call them in the software industry) were developed. One of the most noticeable to the public was changing the shape of the engine intakes from circular to oval, the better to clear the ground.

With the 737 Max, the situation became critical. The engines on the original 737 had a fan diameter (that of the intake blades on the engine) of just 100 centimeters (40 inches); those planned for the 737 Max have 176 cm. That's a

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the wing. However, doing so also meant that the centerline of the engine's thrust changed. Now, when the pilots applied power to the engine, the aircraft would have a significant propensity to "pitch up," or raise its nose.

The angle of attack is the angle between the wings and the airflow over the wings. Think of sticking your hand out of a car window on the highway. If your hand is level, you have a low angle of attack; if your hand is pitched up, you have a high angle of attack. When the angle of attack is great enough, the wing enters what's called an aerodynamic stall. You can feel the same thing with your hand out the window: As you rotate your hand, your arm wants to move up like a wing more and more until you stall your hand, at which point your arm wants to flop down on the car door.

This propensity to pitch up with power application thereby increased the risk that the airplane could stall when the pilots "punched it" (as my son likes to say). It's particularly likely to happen if the airplane is flying slowly

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I'll say it again: In the 737 Max, the engine nacelles themselves can, at high angles of attack, work as a wing and produce lift. And the lift they produce is well ahead of the wing's center of lift, meaning the nacelles will cause the 737 Max at a high angle of attack to go to a *higher* angle of attack. This is aerodynamic malpractice of the worst kind.

Pitch changes with power changes are common in aircraft. Even my little Cessna pitches up a bit when power is applied. Pilots train for this problem and are used to it. Nevertheless, there are limits to what safety regulators will allow and to what pilots will put up with.

Pitch changes with increasing angle of attack, however, are quite another thing. An airplane approaching an aerodynamic stall cannot, under any circumstances, have a tendency to go further into the stall. This is called “dynamic instability,” and the only airplanes that exhibit that characteristic—fighter jets—are also fitted with ejection seats.

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landing gear.

The airframe, the hardware, should get it right the first time and not need a lot of added bells and whistles to fly predictably. This has been an aviation canon from the day the Wright brothers first flew at Kitty Hawk.

Apparently the 737 Max pitched up a bit too much for comfort on power application as well as at already-high angles of attack. It violated that most ancient of aviation canons and probably violated the certification criteria of the U.S. Federal Aviation Administration. But instead of going back to the drawing board and getting the airframe hardware right (more on that below), Boeing relied on something called the “Maneuvering Characteristics Augmentation System,” or MCAS.

Boeing’s solution to its hardware problem was software.

I will leave a discussion of the corporatization of the aviation

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modifying the airframe to accommodate the larger engines. Such an airframe modification would have meant things like longer landing gear (which might not then fit in the fuselage when retracted), more wing dihedral (upward bend), and so forth. All of those hardware changes would be horribly expensive.



“Everything about the design and manufacture of the Max was done to preserve the myth that ‘it’s just a 737.’ Recertifying it as a new aircraft would have taken years and millions of dollars. In fact, the pilot licensed to fly the 737 in 1967 is still licensed to fly all subsequent versions of the 737.”

What’s worse, those changes could be extensive enough to require not only that the FAA recertify the 737 but that Boeing

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is just a 737, and any pilot who has flown other 737s can fly a 737 Max without expensive training, without recertification, without another type of rating. Airlines—Southwest is a prominent example—tend to go for one “standard” airplane. They want to have one airplane that all their pilots can fly because that makes both pilots and airplanes fungible, maximizing flexibility and minimizing costs.

It all comes down to money, and in this case, MCAS was the way for both Boeing and its customers to keep the money flowing in the right direction. The necessity to insist that the 737 Max was no different in flying characteristics, no different in systems, from any other 737 was the key to the 737 Max’s fleet fungibility. That’s probably also the reason why the documentation about the MCAS system was kept on the down-low.

Put in a change with too much visibility, particularly a change to the aircraft’s operating handbook or to pilot training, and someone—probably a pilot—would have nixed it and said.

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and rotating it. It turns out that sophisticated aircraft have what is essentially the mechanical equivalent of a hand out the window: the angle-of-attack sensor.

You may have noticed this sensor when boarding a plane. There are usually two of them, one on either side of the plane, and usually just below the pilot's windows. Don't confuse them with the pitot tubes (we'll get to those later). The angle-of-attack sensors look like wind vanes, whereas the pitot tubes look like, well, tubes.

Angle-of-attack sensors look like wind vanes because that's exactly what they are. They are mechanical hands designed to rotate in response to changes in that angle of attack.

The pitot tubes measure how much the air is “pressing” against the airplane, whereas the angle-of-attack sensors measure what direction that air is coming from. Because they measure air pressure, the pitot tubes are used to determine the aircraft's speed through the air. The angle-of-attack

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instruments on the copilot's side. That gives a state of natural redundancy in instrumentation that can be easily cross-checked by either pilot. If the copilot thinks his airspeed indicator is acting up, he can look over to the pilot's airspeed indicator and see if it agrees. If not, both pilot and copilot engage in a bit of triage to determine which instrument is profane and which is sacred.

Long ago there was a joke that in the future planes would fly themselves, and the only thing in the cockpit would be a pilot and a dog. The pilot's job was to make the passengers comfortable that someone was up front. The dog's job was to bite the pilot if he tried to touch anything.

On the 737, Boeing not only included the requisite redundancy in instrumentation and sensors, it also included redundant flight computers—one on the pilot's side, the other on the copilot's side. The flight computers do a lot of things, but their main job is to fly the plane when commanded to do so and to

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plane down when the system thinks the plane might exceed its angle-of-attack limits; it does so to avoid an aerodynamic stall. Boeing put MCAS into the 737 Max because the larger engines and their placement make a stall more likely in a 737 Max than in previous 737 models.

When MCAS senses that the angle of attack is too high, it commands the aircraft's trim system (the system that makes the plane go up or down) to lower the nose. It also does something else: Indirectly, via something Boeing calls the "Elevator Feel Computer," it pushes the pilot's control columns (the things the pilots pull or push on to raise or lower the aircraft's nose) downward.

In the 737 Max, like most modern airliners and most modern cars, everything is monitored by computer, if not directly controlled by computer. In many cases, there are no actual mechanical connections (cables, push tubes, hydraulic lines) between the pilot's controls and the things on the wings, rudder, and so forth that actually make the plane move. And.

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about what is going on. In the old days, when cables connected the pilot's controls to the flying surfaces, you had to pull up, hard, if the airplane was trimmed to descend. You had to push, hard, if the airplane was trimmed to ascend. With computer oversight there is a loss of natural sense in the controls. In the 737 Max, there is no real "natural feel."

True, the 737 does employ redundant hydraulic systems, and those systems do link the pilot's movement of the controls to the action of the ailerons and other parts of the airplane. But those hydraulic systems are powerful, and they do not give the pilot direct feedback from the aerodynamic forces that are acting on the ailerons. There is only an artificial feel, a feeling that the computer wants the pilots to feel. And sometimes, it doesn't feel so great.

When the flight computer trims the airplane to descend, because the MCAS system thinks it's about to stall, a set of motors and jacks push the pilot's control columns forward. It turns out that the Elevator Feel Computer can put a *lot* of

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THE ANTISTALL SYSTEM DEPENDED CRUCIALLY ON sensors that are installed on each side of the airliner—but the system consulted only the sensor on one side. NOREBB0.COM

Indeed, not letting the pilot regain control by pulling back on the column was an explicit design decision. Because if the pilots could pull up the nose when MCAS said it should go down, why have MCAS at all?

MCAS is implemented in the flight management computer, even at times when the autopilot is turned off, when the pilots think they are flying the plane. In a fight between the flight management computer and human pilots over who is in charge, the computer will bite humans until they give up and (literally) die.

Finally, there's the need to keep the very existence of the

MCAS system a secret. The FAA's investigation of the crash found that the FAA had been told that the system was a "bug" and not a "feature."

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code. And that's where things get dangerous.

Those lines of code were no doubt created by people at the direction of managers. Neither such coders nor their managers are as in touch with the particular culture and mores of the aviation world as much as the people who are down on the factory floor, riveting wings on, designing control yokes, and fitting landing gears. Those people have decades of institutional memory about what has worked in the past and what has not worked. Software people do not.

In the 737 Max, only one of the flight management computers is active at a time—either the pilot's computer or the copilot's computer. And the active computer takes inputs *only* from the sensors on its own side of the aircraft.

When the two computers disagree, the solution for the humans in the cockpit is to look across the control panel to see what the other instruments are saying and then sort it out. In the Boeing system, the flight management computer

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alternates from one extreme environment to another, vibrating and shaking all the way—the flight management computer just believes it.

It gets even worse. There are several other instruments that can be used to determine things like angle of attack, either directly or indirectly, such as the pitot tubes, the artificial horizons, etc. All of these things would be cross-checked by a human pilot to quickly diagnose a faulty angle-of-attack sensor.

In a pinch, a human pilot could just look out the windshield to confirm visually and directly that, no, the aircraft is not pitched up dangerously. That's the ultimate check and should go directly to the pilot's ultimate sovereignty. Unfortunately, the current implementation of MCAS denies that sovereignty. It denies the pilots the ability to respond to what's before their own eyes.

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sees with his own eyes and what he does when he desperately tries to pull back on the robotic control columns that are biting him, and his passengers, to death.

In the old days, the FAA had armies of aviation engineers in its employ. Those FAA employees worked side by side with the airplane manufacturers to determine that an airplane was safe and could be certified as airworthy.

As airplanes became more complex and the gulf between what the FAA could pay and what an aircraft manufacturer could pay grew larger, more and more of those engineers migrated from the public to the private sector. Soon the FAA had no in-house ability to determine if a particular airplane's design and manufacture were safe. So the FAA said to the airplane manufacturers, "Why don't you just have your people tell us if your designs are safe?"

The airplane manufacturers said, "Sounds good to us." The FAA said. "And sav hi to Joe. we miss him."

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Now this is not quite as sinister a conflict of interest as it sounds. It is in nobody's interest that airplanes crash. The industry absolutely relies on the public trust, and every crash is an existential threat to the industry. No manufacturer is going to employ DERs that just pencil-whip the paperwork. On the other hand, though, after a long day and after the assurance of some software folks, they might just take their word that things will be okay.

It is astounding that no one who wrote the MCAS software for the 737 Max seems even to have raised the possibility of using multiple inputs, including the opposite angle-of-attack sensor, in the computer's determination of an impending stall. As a lifetime member of the software development fraternity, I don't know what toxic combination of inexperience, hubris, or lack of cultural understanding led to this mistake.

But I do know that it's indicative of a much deeper problem. The people who wrote the code for the original MCAS system

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Max. That is big strike No. 1. Boeing then tried to mask the 737's dynamic instability with a software system. Big strike No. 2. Finally, the software relied on systems known for their propensity to fail (angle-of-attack indicators) and did not appear to include even rudimentary provisions to cross-check the outputs of the angle-of-attack sensor against other sensors, or even the other angle-of-attack sensor. Big strike No. 3.

None of the above should have passed muster. None of the above should have passed the "OK" pencil of the most junior engineering staff, much less a DER.

That's not a big strike. That's a political, social, economic, and technical sin.

It just so happens that, during the timeframe between the first 737 Max crash and the most recent 737 crash, I'd had the occasion to upgrade and install a brand-new digital autopilot in my own aircraft. I own a 1979 Cessna 172, the most

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components, including redundant flight computers (dual Garmin G5s) and a sophisticated communication “bus” (a Controller Area Network bus) that lets all the various components talk to one another, irrespective of where they are located in my plane. A CAN bus derives from automotive “drive by wire” technology but is otherwise very similar in purpose and form to the various ARINC buses that connect the components in the 737 Max.

My autopilot also includes electric pitch trim. That means it can make the same types of configuration changes to my 172 that the flight computers and MCAS system make to the 737 Max. During the installation, after the first 737 Max crash, I remember remarking to a friend that it was not lost on me that I was potentially adding a hazard similar to the one that brought down the Lion Air crash.

Finally, my new autopilot also implements “envelope protection,” the envelope being the graph of the performance limitations of an aircraft. If my Cessna is *not* being flown by

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As you can see, the similarities between my \$20,000 autopilot and the multimillion-dollar autopilot in every 737 are direct, tangible, and relevant. What, then, are the differences?

For starters, the installation of my autopilot required paperwork in the form of a “Supplemental Type Certificate,” or STC. It means that the autopilot manufacturer and the FAA both agreed that my 1979 Cessna 172 with its (Garmin) autopilot was so significantly different from what the airplane was when it rolled off the assembly line that it was *no longer the same Cessna 172*. It was a different aircraft altogether.

In addition to now carrying a new (supplemental) aircraft-type certificate (and certification), my 172 required a very large amount of new paperwork to be carried in the plane, in the form of revisions and addenda to the aircraft operating manual. As you can guess, most of those addenda revolved around the autopilot system.

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There are instructions on how to detect when the system malfunctions *and how to disable the system, immediately*. Disabling the system means pulling the autopilot circuit breaker; instructions on how to do that are strewn throughout the documentation, repeatedly. Every pilot who flies my plane becomes intimately aware that it is *not* the same as any other 172.

This is a big difference between what pilots who want to fly my plane are told and what pilots stepping into a 737 Max are (or were) told.

Another difference is between the autopilots in my system and that in the 737 Max. All of the CAN bus–interconnected components constantly do the kind of instrument cross-check that human pilots do and that, apparently, the MCAS system in the 737 Max does not. For example, the autopilot itself has a self-contained attitude platform that checks the attitude information coming from the G5 flight computers. If there is a

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planes. In my 172, there are still cables linking the controls to the flying surfaces. The computer has to press on the same things that I have to press on—and its strength is nowhere near as great as mine. So even if, say, the computer thought that my plane was about to stall when it wasn't, I can easily overcome the computer.

In my Cessna, humans still win a battle of the wills every time. That used to be a design philosophy of every Boeing aircraft, as well, and one they used against their archrival Airbus, which had a different philosophy. But it seems that with the 737 Max, Boeing has changed philosophies about human/machine interaction as quietly as they've changed their aircraft operating manuals.

The 737 Max saga teaches us not only about the limits of technology and the risks of complexity, it teaches us about our real priorities. Today, safety doesn't come first—money comes first, and safety's only utility in that regard is in helping to

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are notoriously hard to fix. And by hard, I mean expensive. Software defects, on the other hand, are easy and cheap to fix. All you need to do is post an update and push out a patch. What's more, we've trained consumers to consider this normal, whether it's an update to my desktop operating systems or the patches that get posted automatically to my Tesla while I sleep.

Back in the 1990s, I wrote an article comparing the relative complexity of the Pentium processors of that era, expressed as the number of transistors on the chip, to the complexity of the Windows operating system, expressed as the number of lines of code. I found that the complexity of the Pentium processors and the contemporaneous Windows operating system was roughly equal.

That was the time when early Pentiums were affected by what was known as the FDIV bug. It affected only a tiny fraction of Pentium users. Windows was also affected by similar defects, also affecting only fractions of its users.

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\$800 million in today's money.

I believe the relative ease—not to mention the lack of tangible cost—of software updates has created a cultural laziness within the software engineering community. Moreover, because more and more of the hardware that we create is monitored and controlled by software, that cultural laziness is now creeping into hardware engineering—like building airliners. Less thought is now given to getting a design correct and simple up front because it's so easy to fix what you didn't get right later.

Every time a software update gets pushed to my Tesla, to the Garmin flight computers in my Cessna, to my Nest thermostat, and to the TVs in my house, I'm reminded that none of those things were complete when they left the factory—because their builders realized they didn't have to be complete. The job could be done at any time in the future with a software update.

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add-on board for the computer certified, while software certifications were nil (other than “cannot run on Windows,” “must be written in C++”). This was, admittedly, nearly 10 years ago, and I hope that things have changed since.”

Boeing is in the process of rolling out a set of software updates to the 737 Max flight control system, including MCAS. I don’t know, but I suspect that those updates will center on two things:

1. Having the software “cross-check” system indicators, just as a human pilot would. Meaning, if one angle-of-attack indicator says the plane’s about to stall, but the other one says it’s not so, at least hold off judgment about pushing the nose down into the dirt and maybe let a pilot or two know you’re getting conflicting signals.
2. Backing off on the “shoot first, ask questions later”

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MICAS design. And, when they were not, I do not know or understand what part of the DER process failed to catch the fundamental design defect.

But I suspect that it all has to do with the same thing that brought us from Boeing's initial desire to put larger engines on the 737 and to avoid having to internalize the cost of those larger engines—in other words, to do what every child is taught is impossible: get a free lunch.

The emphasis on simplicity comes from the work of Charles Perrow, a sociologist at Yale University whose 1984 book, *Normal Accidents: Living With High-Risk Technologies*, tells it all in the very title. Perrow argues that system failure is a normal outcome in any system that is very complex and whose components are “tightly bound”—meaning that the behavior of one component immediately controls the behavior of another. Though such failures may seem to stem from one or another faulty part or practice, they must be seen as inherent in the system itself. They are “normal” failures.

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and then repatch such a system in an attempt to make it safer can end up making it less safe.

This is the root of the old engineering axiom “Keep it simple, stupid” (KISS) and its aviation-specific counterpart: “Simplify, then add lightness.”

The original FAA Eisenhower-era certification requirement was a testament to simplicity: Planes should not exhibit significant pitch changes with changes in engine power. That requirement was written when there was a direct connection between the controls in the pilot’s hands and the flying surfaces on the airplane. Because of that, the requirement—when written—rightly imposed a discipline of simplicity on the design of the airframe itself. Now software stands between man and machine, and no one seems to know exactly what is going on. Things have become too complex to understand.

I cannot get the parallels between the 737 Max and the space shuttle Challenger out of my head. The Challenger accident,

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were weighed, the process was followed, and a majority consensus was to launch. And seven people died.

In the 737 Max case, the rules were also followed. The rules said you couldn't have a large pitch-up on power change and that an employee of the manufacturer, a DER, could sign off on whatever you came up with to prevent a pitch change on power change. The rules didn't say that the DER couldn't take the business considerations into the decision-making process. And 346 people are dead.

It is likely that MCAS, originally added in the spirit of increasing safety, has now killed more people than it could have ever saved. It doesn't need to be "fixed" with more complexity, more software. It needs to be removed altogether.

An earlier version of this article was cited in EE Times.

Editor's note: This story was updated on 21 April to clarify that the MCAS pushes the airliner's nose by means of the "Elevator

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